Saeed Chamani

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Education

Iran University of Science and Technology Sep 2022 – Sep 2025
M.Sc in Bioelectric Engineering
- GPA: 18.70/20

Tabriz University Sep 2017 – Sep 2022
B.Sc in Electrical Engineering (Electronics)
- GPA: 15.68/20

Research Interests

- Machine/Deep Learning
- Generative AI with LLMs
- Image Processing
- Signal Processing
- Computer Vision and Intelligent Video Analytics
- Brain Computer Interfaces
- EEG Analysis and Computational Neuroscience
- Optimization Techniques and Algorithms

Publications

Published

Multi-Attention Enhanced Encoder-Decoder Network with Hybrid Transformer Bottleneck for Echocardiography Image Segmentation

Saeed Chamani, Hamid Behnam

Scientific Reports, 2026

DOI: 10.1038/s41598-026-60699-0 | Nature

Enhancing skin cancer diagnosis using late discrete wavelet transform and new swarm-based optimizers

Ramin Mousa, ***Saeed Chamani***, Mohammad Morsali, Seyyed Mohammad Ojagh Kazzazi, Parsa Hatami, Soroush Sarabi

Machine Learning with Applications, 2025

DOI: 10.1016/j.mlwa.2025.100811 | ScienceDirect

Experience

Research Assistant
- Image Processing Laboratory – Prof. Hamid Behnam Sep 2022–Sep 2025
- Electronic Device Laboratory – Prof. Ali Rostami Sep 2021–Jan 2022

Teaching Assistant
- Industrial Electronics – Prof. M. Sabahi Sep 2019-Dec 2019

Relative Projects

- Advanced Traffic Video Analytics using YOLOv8, DeepSORT, ByteTrack, and Ultralytics
- Automatic Number Plate Recognition (ANPR) using YOLOv8, EasyOCR, DeepSORT
- Autonomous Driving Perception with YOLOv8 on KITTI Dataset
- Preprocessing and classification of EEG signals from schizophrenia datasets using machine and deep learning methods
- Sleep stage prediction using neural networks based on EEG signal analysis
- Implementation of U-Net Architecture with Popular Pretrained Encoders for Enhanced Feature Extraction
- Implementation of 2D UNETR, Swin-Unet, U-Net3+, and U-Net++ for Brain Tumor Segmentation
- Arrangement and preprocessing of 3D medical images (PET/CT, CT, and MRI)
- Implement 3D Deep Network for MRI image segmentation
- Comparative Analysis of Seven Machine Learning Models for Classifying the Indian Liver Patient Dataset
- Morphological Biomedical Image Processing Using MATLAB
- Noise Detection and Reduction in Biomedical Images Using Diverse Methods: Wiener Filter, Wavelet-Based Denoising, DCT, BM3D, Sparse Coding, Total Variation, Diffusion, NLM, ICA, and Bilateral Filtering
- Design and Implementation of Triple Motor Driver System Using VHDL
- Arduino-Based System for Flame Detection and Identification Across Multiple Wavelengths
- Head Model Construction Using FEM and BEM Methods in the FieldTrip Toolbox
- Source Reconstruction of Discrete and Distributed Brain Activity Using Beamforming Techniques

Technical and Language skills

Programming Languages & Tools

Python (TensorFlow, PyTorch, Keras), MATLAB, MNE Python, EEGLAB, git, LaTeX, MySQL, Docker, Verilog, Microsoft Office (Visio, Word, PowerPoint, Excel), Xilinx ISE, Proteus, Field II, Ultrasim

Language skill

Turkish / Azerbaijani (Native proficiency), Persian (Native proficiency), English (Professional working proficiency)

Additional Experience And Awards

- **Top-ranked student** among all M.Sc. students in Bioelectric Engineering at Iran University of Science and Technology
- **Ranked Under 2%** among 180,000 participants, National Universities Entrance Exam (Konkour) for Bachelor's Degree
- **Ranked Under 1%** among 15,000 participants, National Universities Entrance Exam (Konkour) for Master's Degree

References

1. **Dr. Hamid Behnam**
   Electrical Engineering Department of Iran University of Science and Technology
   Associate Professor
   behnam@iust.ac.ir
2. **Dr. Sirous Toofan**
   Electrical & Computer Engineering Department of Tabriz University
   Associate Professor
   s.toofan@tabrizu.ac.ir

Relevant Courses: Brain Computer Interfaces (18.9/20) | Wavelet Applications in Signal and Image Processing (19.8/20) | Pattern Recognition (17.3/20) | Ultrasound Applications in Biomedical Engineering (20/20) | Functional Brain Imaging Systems (18.3/20) | Signals and Systems (19.5/20) | Digital Signal Processing (DSP) (19/20) | Pulse and Digital Circuits (19/20) | Digital Systems Design (ASIC, FPGA) (19/20)